SABER-PID: Source-Isolated Qualification and Cost-Aware Operation of Vision–Language Models for P&ID Tag Retrieval

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Abstract

Reliable engineering-document retrieval requires evidence that predicted tags come from the requested drawing and an operating mode matched to the cost of misses and false candidates. We introduce SABER-PID, a source-isolated, answer-hidden workflow that qualifies vision–language models (VLMs) for candidate-tag retrieval from piping and instrumentation diagrams (P&IDs) and then converts frozen OCR–VLM outputs into a cost-aware decision. On 100 unseen PIDQA drawings, Qwen3-VL-8B reached strict tag F1 0.5549 with the correct drawing, versus 0.0062 with a source-shuffled drawing and 0 without an image. At a common output cap, increasing the visual-input budget produced a +0.5250 F1 effect (95% source-bootstrap interval [0.4224, 0.6293]). Across 230 source-disjoint drawings, clean, JPEG, mild-blur, and downsample-and-restore conditions retained correct-minus-shuffled effects of 0.548–0.575. A closest-safe 54-tile InternVL3.5-8B configuration independently retained a +0.4715 requested-drawing effect ([0.4064, 0.5332]). On 35 images from 26 public DEXPI test cases, Qwen reached F1 0.9057, while both cross-case shuffled and no-image controls scored zero. OCR and VLM errors were complementary: union achieved recall 0.7040 and F1 0.6339, whereas intersection achieved precision 0.9907. Under a linear missed-tag and false-candidate loss, the exact lower envelope selects intersection, joined OCR, OCR-first, or union as the relative miss cost increases. SABER-PID therefore provides a reproducible qualification-to-operation path for configurable P&ID tag retrieval under the evaluated drawing families and budgets.

Keywords: P&ID; engineering document intelligence; vision–language model; tag retrieval; counterfactual evaluation; cost-sensitive decision

1 Introduction

Piping and instrumentation diagrams (P&IDs) are high-density engineering documents whose equipment and instrument identifiers link drawings to asset registers, maintenance records, and process information. Recovering a useful candidate set of tags can support search, indexing, migration, and deterministic downstream validation before complete diagram digitization is available. Vision–language models (VLMs) make this task accessible without training a dedicated detector for every drawing convention, but a plausible answer is not yet engineering evidence: it may follow task priors, repeated drawing identity, or an unrelated image, and small text may disappear under an inadequate visual-input budget.

This motivates two linked decisions. First, does a frozen model actually use the *requested* drawing at a declared visual and output budget? Second, once that capability is established, which transparent retrieval mode best fits a project's relative cost of missed tags, false candidates, and review capacity? Treating only the first decision produces a benchmark result without an operating rule; treating only the second can optimize outputs that were never shown to be drawing-dependent.

P&ID research has advanced symbol and text recognition, line extraction, graph conversion, and end-to-end digitization [1, 2, 3, 4]. Recent multimodal and retrieval systems extend this direction through structured P&ID representations, graph-based interaction, and process-graph construction [7, 8, 9]. PIDQA provides 64,000 graph-derived questions from 500 synthetic source sheets [6], making controlled VLM evaluation possible. However, many question records share a drawing, so question-random splitting can expose source-specific information on both sides of a split. Document-VQA work also shows that surface-form correctness and document groundedness are different properties [12, 14]. An engineering qualification therefore needs drawing-level isolation and direct interventions on visual evidence.

We call the workflow *SABER-PID*. It isolates drawings by source, hides answers from inference, records budgets, intervenes on visual evidence, and reports each task separately. SABER-PID is an engineering measurement and operating framework rather than a new model architecture. Figure 1 gives its five-stage path. The contributions are:

1. a source-level qualification contract that intervenes on requested, source-shuffled, and absent images while keeping references outside inference;
2. matched-budget, mild-quality, cross-model, and second-family transfer tests with source- or logical-case-cluster uncertainty;
3. a deterministic family of OCR–VLM set and fallback modes with candidate workload, precision–recall, and bootstrap stability reporting; and
4. an explicit cost-sensitive decision rule that converts fixed TP/FP/FN counts into a selectable engineering operating mode.

The central finding is constructive and bounded: requested-drawing tag evidence persists across the qualified Qwen setting, declared mild image shifts, a closer-budget InternVL setting, and a public DEXPI family; transparent OCR–VLM modes then expose useful precision–coverage choices. The work does not claim topology reconstruction, arbitrary-document robustness, or autonomous field deployment.

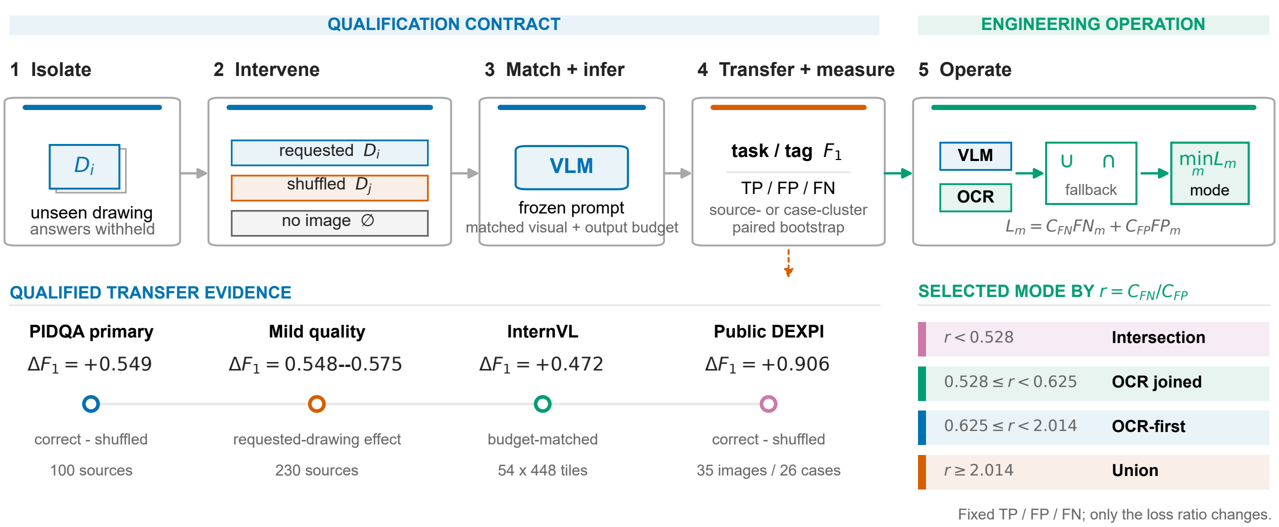

Figure 1: SABER-PID qualification-to-operation workflow. The upper backbone follows drawing-level isolation, visual-evidence intervention, budget-matched inference, transfer measurement, and deterministic OCR–VLM operation. The lower rails summarize retained transfer evidence and map the cost ratio $r = C_{FN}/C_{FP}$ to the minimum-loss operating mode. All values are deterministic compositions of frozen score reports.

2 Related work and research gap

2.1 P&ID extraction, structured representations, and interaction

Classical P&ID digitization pipelines decompose the problem into recognition of symbols, text, lines, and associations before graph construction [1, 2]. High-density industrial studies demonstrate the value of specialized symbol and text recognizers [3], while recent work integrates symbol, text, and pipeline detection in high-resolution drawings [4]. Semantic normalization remains necessary because engineering identifiers and metadata are often non-standardized [5].

The newest P&ID language-model work pursues complementary outputs. Talking like P&IDs and ChatP&ID study multimodal or graph-mediated interaction with engineering diagrams [7, 8]; Zhu et al. target conversion of drawing content into process graphs [9]. These systems highlight the value of structured engineering representations. Our question is different: before a raw-image VLM output is admitted to such a pipeline, can its candidate tags be attributed to the requested drawing under a declared budget, and can its operating trade-off be made explicit?

2.2 Grounding, counterfactuals, and grouped evaluation

DocVQA and ChartQA established document question answering as a combined text, layout, and reasoning problem [12, 13]. Grounding-aware evaluation asks whether the answer is supported by the input document rather than only whether its surface form matches a reference [14]. Counterfactual samples and visual contrasts expose shortcut dependence and unsupported VLM outputs [15, 16, 17]. Separately, multiple-source cross-validation and leakage-aware evaluation require related observations to remain in the same partition [10, 11].

SABER-PID joins these principles in an engineering decision contract. The drawing is the split and resampling unit; references are unavailable during inference; requested, shuffled, and absent images isolate visual-source dependence; input and output budgets are recorded separately; and outputs are reported as task-specific candidate sets rather than an aggregate QA score. The resulting novelty is the complete path from source-specific qualification to cost-aware operation, not another prompt comparison or recognition model.

3 Materials and methods

3.1 Datasets, source identity, and answer isolation

PIDQA contains 64,000 questions from 500 Dataset-PID sheets, equally divided among count, spatial-count, connectivity, and value-retrieval tasks [6]. Primary Set B contains one deterministically selected record per source and task for 100 source sheets (400 records); strict tag metrics use the 100 value-retrieval records. Public identifiers are ranked by SHA-256. Answers, graph queries, and predictions never enter selection. Each model-facing manifest contains only instance ID, source ID, task, question, image path, and public template fields. References reside in a scorer-only store. Automated schema and leakage checks cover all reported public manifests and raw outputs; immutable artifacts are identified by SHA-256 in the release manifest.

The external branch uses the CC BY 4.0 DEXPI Public Example PIDs repository at frozen commit `a23d61e2e089eb2ca464cd552f9ae580a2785963` [21]. An automated, output-blind audit accepts same-directory image/XML pairs with identical stems and retains a structured tag only when the XML graphics declare it as visible text. Duplicate image hashes are removed, and logical-case coverage is maximized before vendor variants are added. The resulting set contains 35 images from 26 logical test cases and 65 prefix-conditioned tag questions. Public manifests again contain no answer-bearing field.

3.2 Qualification conditions and frozen instruments

The primary instrument is Qwen3-VL-8B-Instruct with a pre-specified prompt and greedy decoding [19]. It receives a maximum image side of 768 or 3072 and an independently labelled output cap of 192 or 512 tokens. The matched-budget contrast fixes the output cap at 512. Source-shuffled images use a deterministic Sattolo derangement across the 100 drawings, with no fixed point. The no-image condition preserves model, prompt, question, decoder, and output cap while removing image content.

The frozen extension crosses Qwen3-VL-8B, Qwen3-VL-32B, and InternVL3.5-8B [20] with Set B and two additional pairwise source-disjoint 65-drawing subsets, two pre-existing prompts, and correct, source-shuffled, and no-image conditions. Qwen uses 3072-side images and a 512-token cap. The initial InternVL comparison uses at most 12 realised 448-pixel tiles and the same output cap. Every pre-specified cell is retained; no best-prompt or best-subset selection is performed.

Two focused matrices retain the qualified 3072/512 Qwen setting. The quality matrix crosses all three source-disjoint subsets (230 drawings in total) with clean images, JPEG quality 70, Gaussian blur radius 1, and 0.75× downsample-and-restore; every condition has a paired source-shuffled control. The closer-budget model comparison uses InternVL with 54 non-overlapping 448-pixel tiles in a 9×6 white-letterbox canvas, no thumbnail, and the same 512-token cap. Its recorded input tensor has 9.63% fewer elements than the Qwen processor. This is a closest-safe gross-budget comparison, not architecture or information-throughput equivalence.

The DEXPI matrix uses Qwen at 3072/512 under correct, cross-case shuffled, and no-image conditions, plus the same frozen OCR family. The shuffled manifest is a one-to-one derangement with no fixed point and no mapping within one logical case; among valid derangements, a fixed-seed search minimizes tag-set overlap. Neither image selection nor derangement uses model output.

3.3 OCR–VLM operating modes and cost rule

The OCR comparator uses PaddleOCR 2.8.1 with PaddlePaddle 2.6.2, the English PP-OCRv3 detector and PP-OCRv4 recognizer, full images, no angle classifier, and detector maximum side 3072 [18]. A deterministic vertical prefix–suffix geometry join reconstructs split identifiers using only OCR coordinates and a frozen legal-tag grammar. It never reads a reference.

Let Q_i and O_i denote Qwen and geometry-joined OCR candidate sets for drawing i. Four score-free operations are evaluated: union $Q_i \cup O_i$, intersection $Q_i \cap O_i$, OCR-if-nonempty with Qwen fallback, and Qwen-if-nonempty with OCR fallback. The complete rule family is preserved in the supplement. Union targets coverage, intersection a conservative shortlist, and OCR-first an intermediate operating point. Set B describes component complementarity and the initial operating envelope; later frozen-rule checks, including source-overlap exclusions, are reported as within-family sensitivity rather than external replication.

For an explicit operating choice, false-positive cost is normalized to one and $r = C_{\mathrm{FN}}/C_{\mathrm{FP}}$. Each mode m receives

$$L_m(r) = r\, FN_m + FP_m.$$

The exact lower envelope of these linear functions yields the point-estimate decision intervals. On a predeclared log-spaced ratio grid, 10,000 source-cluster resamples recompute all four losses and split tie mass equally. This scorer changes only the selected operating mode; it does not alter a prediction.

3.4 Metrics, uncertainty, and workload

For source i, reference and prediction sets are T_i and P_i, with $TP_i = |T_i \cap P_i|$, $FP_i = |P_i \setminus T_i|$, and $FN_i = |T_i \setminus P_i|$. Pooled estimators sum counts before computing precision, recall, and

$$F1_{\mathrm{micro}} = \frac{2\sum_i TP_i}{2\sum_i TP_i + \sum_i FP_i + \sum_i FN_i}.$$

Source-macro metrics average per-source values. Exact-set accuracy is the fraction with normalized set equality. Seven Set B sources have empty references; the supplement reports both the declared empty-set convention and a non-empty-reference macro sensitivity.

All intervals are 95% percentile intervals from 10,000 fixed-seed source- cluster bootstrap replicates. Paired comparisons resample identical source indices. DEXPI intervals resample its 26 logical cases so that all questions and vendor variants from one case move together. Quality effects use a paired difference-in-differences: a degraded correct-minus-shuffled contrast minus the clean contrast. Workload is the per-drawing candidate count $|P_i|$ and false-candidate count $|P_i \setminus T_i|$, summarized by median and interquartile range (IQR).

4 Results

4.1 Requested-drawing qualification on unseen PIDQA sources

The primary Qwen setting produces 192 true positives, 152 false positives, and 156 false negatives on 100 unseen drawings. Pooled precision, recall, and strict tag F1 are 0.5581, 0.5517, and 0.5549. Source-macro F1 is 0.5810 ([0.5118, 0.6501]); excluding seven empty-reference sources yields 0.6139. When the image is replaced by a different source drawing, F1 falls to 0.0062; without an image it is zero. Correct minus source-shuffled F1 is +0.5487 ([0.4729, 0.6239]), and correct minus no-image F1 is +0.5549 ([0.4799, 0.6322]). The useful tag signal therefore follows the requested drawing rather than the question template or arbitrary P&ID pixels.

Visual detail is also consequential. At a common 512-token output cap, 3072-side Qwen reaches F1 0.5253 and 768-side Qwen reaches 0.0003, a +0.5250 paired effect ([0.4224, 0.6293]). The conclusion is therefore tied to a declared visual-input budget, not to model name alone. Table 1 summarizes the four central qualification and transfer settings.

Table 1: Qualification scorecard across drawing, quality, and model families. Effects are correct-image minus source-shuffled strict tag F1. Intervals use 10,000 grouped bootstrap replicates; the mild-quality row reports ranges over the four frozen conditions.

Setting	Evaluation unit	Correct F1	Shuffled F1	Requested-drawing effect [95% interval]
PIDQA, Qwen primary	100 sources	0.5549	0.0062	+0.5487 [0.4729, 0.6239]
PIDQA mild-quality matrix	230 disjoint sources	0.5605–0.5879	0.0114–0.0125	+0.5482–+0.5754
PIDQA, InternVL 54-tile	230 disjoint sources	0.4846	0.0131	+0.4715 [0.4064, 0.5332]
Public DEXPI, Qwen	26 logical cases / 35 images	0.9057	0.0000	+0.9057 [0.8409, 0.9841]

A separate split diagnostic explains why source isolation matters. Across five pre-specified seeds, an input-retrieval-plus-training-prior diagnostic scores 50.9% under question-random splitting and 31.5% under source-isolated splitting, a mean difference of 19.5 percentage points. Exact-image cache hits create this route only when a drawing crosses the split. This is a deterministic exposure diagnostic, not a claim about the evaluated VLM's internal mechanism. Aggregate QA accuracy is likewise retained as a boundary: correct-image accuracy (0.2875) does not exceed no-image accuracy (0.3025), so the qualified claim is candidate-tag retrieval rather than general P&ID QA.

4.2 Quality robustness and matched-budget model transfer

Across 230 pairwise source-disjoint drawings, correct-image strict tag F1 is 0.5693 on clean images, 0.5836 after JPEG quality 70, 0.5879 after Gaussian blur radius 1, and 0.5605 after 0.75× downsample-and-restore. The paired correct-minus-shuffled effects are +0.5578 ([0.4917, 0.6227]), +0.5722 ([0.5060, 0.6344]), +0.5754 ([0.5101, 0.6356]), and +0.5482 ([0.4763, 0.6134]), respectively. Changes in this requested-drawing effect relative to clean are +0.0143 ([-0.0034, 0.0345]) for JPEG, +0.0176 ([-0.0155, 0.0517]) for blur, and −0.0096 ([-0.0655, 0.0520]) for downsample-and-restore. Thus all declared mild shifts retain a large effect, and no deterioration from clean is detected.

At the closer safe 54-tile InternVL budget, pooled correct-image F1 is 0.4846, versus 0.0131 under source shuffling and zero without an image. Correct minus shuffled is +0.4715 ([0.4064, 0.5332]); correct minus no image is +0.4846 ([0.4198, 0.5463]). The gain over the earlier native 12-tile setting is +0.4681 ([0.4043, 0.5290]). Figure 2 shows both robustness and cross-model recovery. The result supports transfer of requested-drawing dependence under a closer gross visual budget; it does not assert encoder equivalence.

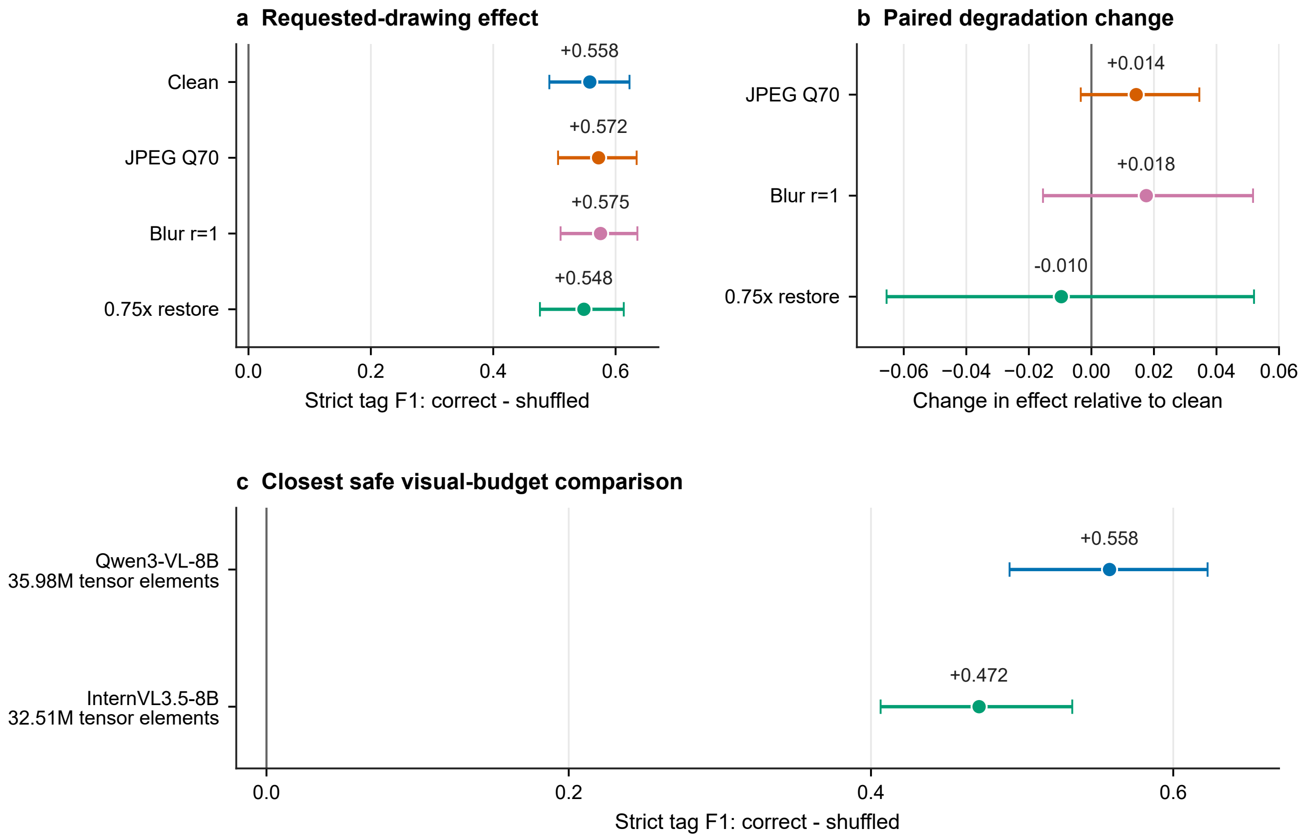

Figure 2: Qualified-budget robustness and cross-model visual-budget recovery. Panel A reports correct-minus-shuffled strict tag F1 under clean and three mild quality shifts. Panel B reports paired change relative to clean. Panel C shows InternVL requested-drawing effects at the closest-safe 54-tile setting. Whiskers are paired 95% source-bootstrap intervals; recorded tensor-element matching does not equalize encoders.

The full pre-specified three-model, three-subset, two-prompt matrix remains informative as a budget boundary. All 36 correct-minus-control intervals have positive lower bounds, including the initial native-budget InternVL cells, but those native correct-image F1 values are only 0.0090–0.0257. Qwen8B and Qwen32B maintain useful magnitudes across subsets and prompts; eight of nine prompt contrasts include zero. Complete cell values and the contrast figure are retained in the supplement rather than used for best-cell selection.

4.3 Transfer to a public DEXPI drawing family

The visibility-qualified DEXPI branch contains 65 questions from 35 images in 26 logical cases. On exact image/XML pairs, Qwen produces 72 true positives, six false positives, and nine false negatives:

precision 0.9231, recall 0.8889, strict tag F1 0.9057, and exact-set accuracy 0.8615. Both cross-case shuffled and no-image conditions produce zero true positives and F1 0. Correct minus shuffled and correct minus no image are both +0.9057, with logical-case bootstrap intervals [0.8409, 0.9841] and [0.8431, 0.9844].

The same frozen full-image OCR family reaches precision 0.9608, recall 0.6049, F1 0.7424, and exact-set accuracy 0.6154. Qwen correct-image minus OCR F1 is +0.1632 ([0.0317, 0.3882]). Figure 3 retains every condition. Because source selection, visibility filtering, image/XML pairing, and derangement are output-blind, this provides transfer evidence to a second licensed public P&ID family. Its curated test-case character and scale do not represent operating-plant diversity.

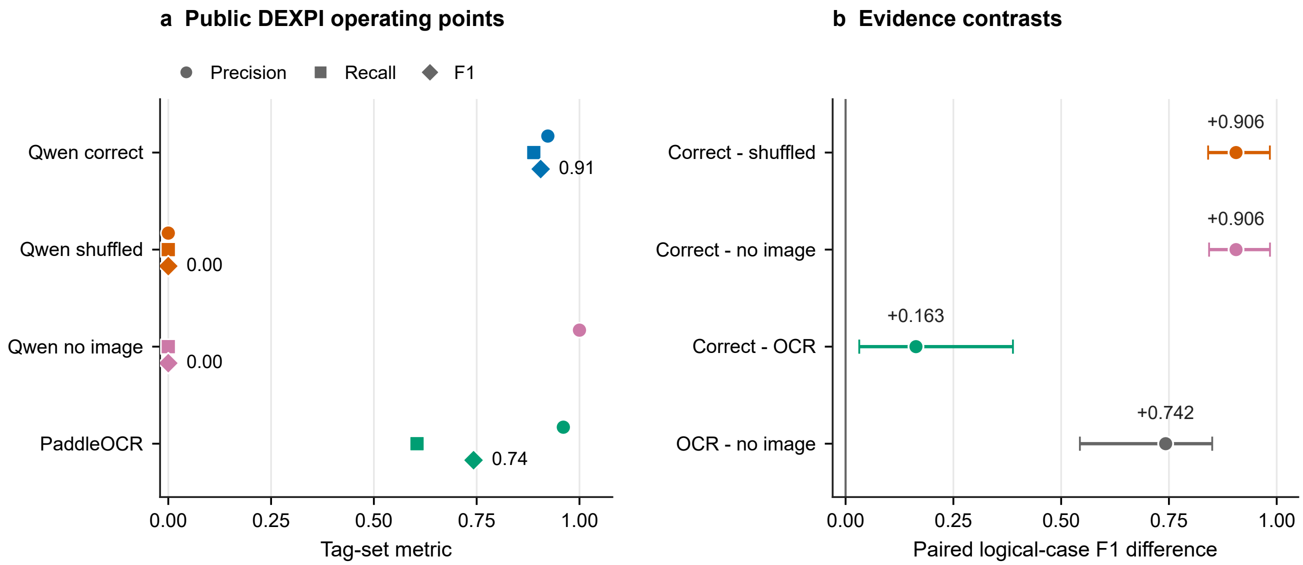

Figure 3: External-family tag retrieval on public DEXPI drawings. Panel A shows precision, recall, and F1 for Qwen correct, cross-case shuffled, no-image, and full-image OCR conditions. Panel B reports logical-case-bootstrap F1 contrasts. All predictions are retained; no result-guided drawing or tag selection is applied.

4.4 Cost-aware operating modes

Qwen and geometry-joined OCR recover complementary evidence on Set B. Of 348 reference tags, 106 are found by both instruments, 86 only by Qwen, 53 only by OCR, and 103 by neither. Among 180 distinct false candidates in their union, only one appears in both. This complementarity produces useful transparent operating points (Table 2). Union recovers 245 tags, reaching recall 0.7040 and F1 0.6339. Intersection returns 107 candidates with only one false candidate, reaching precision 0.9907 at recall 0.3046. OCR-first reaches precision 0.8178, recall 0.5029, and F1 0.6228. Median [IQR] candidates per drawing are 4 [2, 5] for union, 1 [0, 2] for intersection, and 2 [1, 3] for OCR-first.

The observed counts yield the loss lines intersection $242r + 1$, joined OCR $189r + 29$, OCR-first $173r + 39$, and union $103r + 180$. Their exact lower envelope selects intersection for $r < 0.5283$, joined OCR for $0.5283 < r < 0.6250$, OCR-first for $0.6250 < r < 2.0143$, and union for $r > 2.0143$; adjacent modes tie at a boundary. Hence a workflow valuing a miss and a false candidate similarly selects OCR-first, a review-sensitive workflow selects intersection, and a sufficiently miss-sensitive workflow selects union.

Bootstrap stability is strongest away from switch points. Intersection is nearly certain at low r, union at high r, and OCR-first reaches a maximum minimum-loss probability of 0.889 near $r = 1.25$.

Table 2: Transparent PIDQA operating modes on 100 unseen sources. Workload columns are per-drawing median [IQR].

Mode	Engineering role	TP	FP	FN	Precision	Recall	F1	Candidates	False candidates
Qwen	single VLM	192	152	156	0.5581	0.5517	0.5549	3 [1, 4]	0 [0, 2]
Joined OCR	low-false-positive OCR	159	29	189	0.8457	0.4569	0.5933	2 [1, 3]	0 [0, 0]
Intersection	precision-first shortlist	106	1	242	0.9907	0.3046	0.4659	1 [0, 2]	0 [0, 0]
OCR-first	balanced fallback	175	39	173	0.8178	0.5029	0.6228	2 [1, 3]	0 [0, 1]
Union	recall-first coverage	245	180	103	0.5765	0.7040	0.6339	4 [2, 5]	1 [0, 2]

Joined OCR occupies a narrow point-estimate interval and reaches at most 0.347. Figure 4 combines the precision–recall/workload view, exact lower-loss envelope, and source-bootstrap stability; it is a decision aid tied to the evaluated count distribution, not a universal tariff.

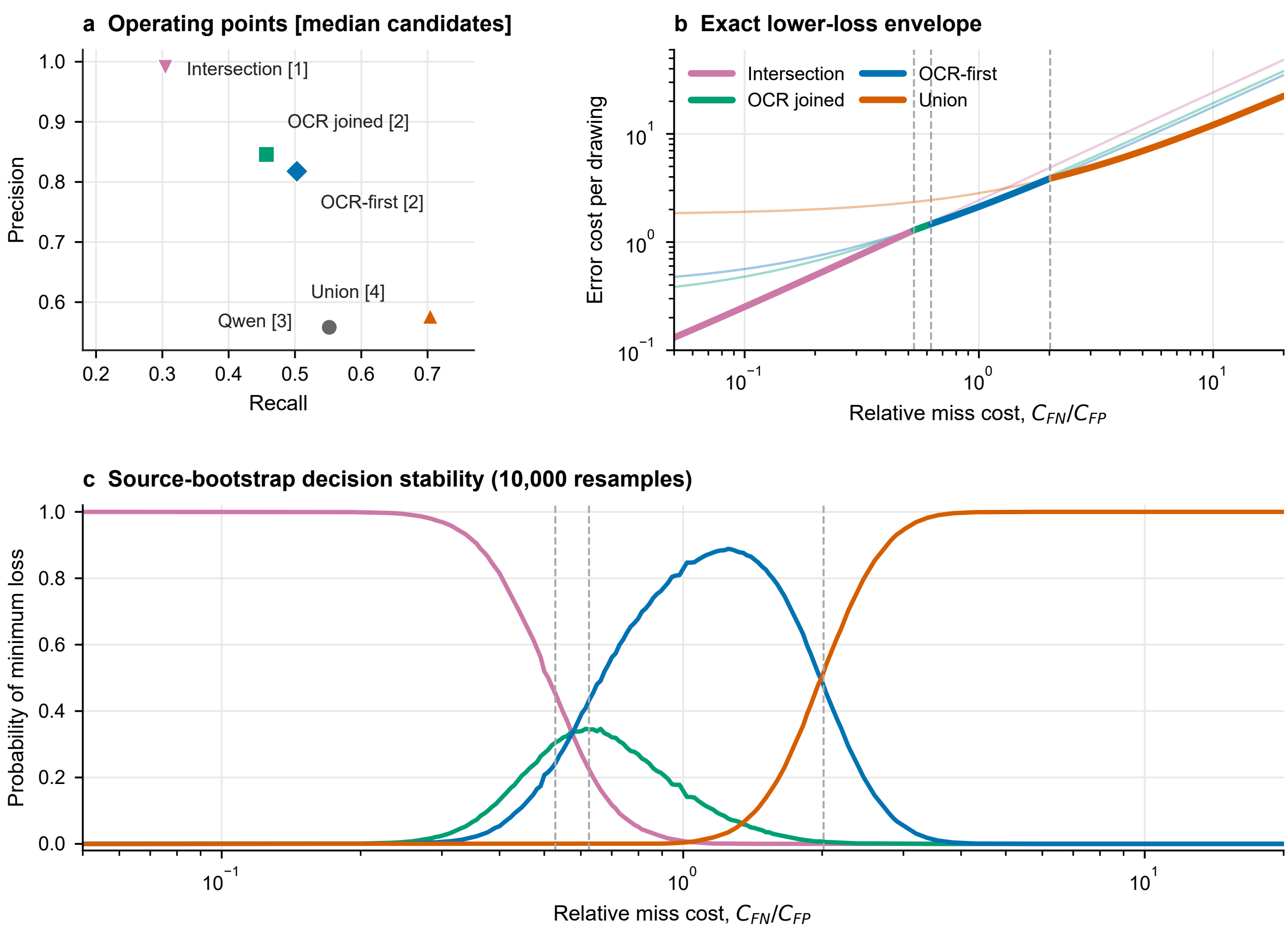

Figure 4: Cost-aware operation from fixed OCR–VLM outputs. Panel A shows precision–recall points with median candidate workload. Panel B highlights the exact lower-loss envelope for $L = rFN + FP$. Panel C gives each mode's probability of minimum loss over 10,000 source-cluster resamples. Dashed lines mark point-estimate switches; scoring changes no prediction.

The deterministic OCR geometry join improves pooled F1 from 0.4567 to 0.5933 by adding 44 true tags and one false candidate while removing 16 false candidates and one true tag. Frozen union checks after source-overlap exclusions retain positive point estimates (+0.0325 and +0.0509 on two strictly disjoint 65-source subsets), although some intervals include zero. These analyses support the mechanism and directional stability of the operating family; the supplement reports every estimator and exclusion.

5 Discussion

5.1 From benchmark score to engineering qualification

The results show why task-level, source-specific evidence can be more useful than aggregate accuracy. Aggregate PIDQA QA accuracy does not distinguish the correct image from no image, yet candidate-tag retrieval shows a +0.5487 correct-minus-shuffled effect and a +0.5250 matched-output-cap visual-budget effect. The qualification becomes stronger, rather than broader, when its conditions are explicit: the signal persists under three mild image interventions, at a closer safe budget in another VLM family, and on a second public P&ID family with zero-F1 counterfactual controls.

The experimental design localizes several otherwise conflated factors. Drawing-level splitting removes an input-recoverable same-source route. Answer-hidden manifests separate inference from scoring. Source shuffling tests whether evidence follows the requested drawing, while no-image controls test whether an image is necessary. Matching the output cap separates visual detail from generation allowance. Recording actual model inputs reveals why a nominally cross-model comparison may instead be a visual-budget comparison. These controls form a reusable engineering evaluation pattern for other dense technical documents.

5.2 Operational value of transparent mode selection

The OCR–VLM results turn qualification into an actionable workflow. A precision-first shortlist can use intersection, an intermediate workflow can use OCR-first, and a coverage-first indexing or discovery task can use union. The cost envelope makes this more than an informal label: once a project declares a relative miss cost, the sample-derived rule selects a mode and the bootstrap panel displays how stable that choice is. Candidate-count and false-candidate distributions simultaneously expose review burden.

These modes are deliberately transparent. They do not require a trained fusion layer, do not use hidden references at prediction time, and do not change component outputs. The same design can therefore be implemented as a deterministic layer around frozen instruments and audited independently. Project-specific identifier grammars or controlled master-data checks can be added after retrieval without changing the evidence claim made here.

5.3 Relationship to broader P&ID intelligence

SABER-PID complements, rather than replaces, symbol detection, topology reconstruction, graph construction, and graph-grounded interaction. Its output is a candidate tag set suitable for indexing or deterministic validation, not an autonomous engineering decision. Structured systems such as ChatP&ID and process-graph pipelines can benefit from an explicit admission test for raw-image evidence; conversely, their topology and schema constraints can provide downstream validation unavailable to tag-only retrieval. The engineering contribution is therefore a qualified interface between raw drawing perception and higher-level document intelligence.

6 Limitations

The main PIDQA experiments use a synthetic 500-sheet family. Drawing-level isolation protects unseen-source evaluation within that family but does not establish performance on proprietary, scanned, revision-marked, handwritten, or severely degraded plant documents. DEXPI adds a licensed external family, but only 35 visibility-qualified images from curated exchange test cases; it is not a real-plant sample. The quality matrix covers one mild severity for JPEG, blur, and resampling and cannot support robustness claims outside those declared transformations.

The primary operating point uses one frozen Qwen3-VL-8B prompt and decoder. The extension includes a 32B Qwen scale, a second prompt, and InternVL3.5-8B, but only one non-Qwen family. The native-to-54-tile InternVL change shows that effective visual budget is not invariant; approximate tensor-element matching does not equalize encoders, architectures, or information throughput. Full native-budget, prompt-sensitivity, aggregate/structural-task, and source- overlap analyses are retained in the supplement and release, including the small native InternVL scores and intervals that cross zero in some strict union sensitivities.

Set B operating modes are descriptive, and frozen-rule checks remain within PIDQA. The cost envelope estimates a rule from benchmark TP/FP/FN counts; real projects must supply their own miss and false-candidate costs and may have a different tag distribution. Candidate workload is benchmark-derived rather than measured human effort. The OCR comparator is a frozen full-image pipeline rather than an optimized industrial recognizer. Finally, tag retrieval does not qualify symbol identity, connectivity, topology, process reasoning, or field deployment.

7 Conclusion

SABER-PID links two decisions that engineering use of VLM outputs requires: whether candidate tags are supported by the requested P&ID and which transparent operating mode fits the intended cost profile. Correct, source-shuffled, and no-image interventions establish a +0.5487 requested-drawing effect on unseen PIDQA sources; matched-output-cap testing shows a +0.5250 visual-budget effect. The evidence persists across declared mild image shifts, a closest-safe InternVL budget, and a public DEXPI family where correct-image F1 reaches 0.9057 and both controls score zero. Complementary OCR–VLM errors then support precision-first, intermediate, and recall-first modes, with an exact cost envelope selecting among them. The result is a reproducible qualification-to-operation workflow for bounded P&ID tag retrieval and a practical template for evaluating multimodal systems in dense engineering documents.

Data and code availability

The submission package contains an independently reproducible archive with answer-isolated public manifests, scorer-only references, immutable raw generation JSONL, deterministic analysis and figure scripts, tests, environment records, and SHA-256 manifests. PIDQA/Dataset-PID source data are covered by the vendored CC0 record; complete image collections are acquired by reference, and model weights are not redistributed. Before submission, the authors will deposit the release archive and replace `[SUBMITTER: ARCHIVE DOI/URL]` with its persistent URL or DOI.

Declaration of competing interest

The submitting authors must insert and approve the applicable competing- interest declaration before upload; no declaration is inferred by the automated revision workflow.

Funding

The submitting authors must insert the applicable funding statement before upload.

Use of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI Codex and ChatGPT Pro to assist analysis-code development, deterministic validation, figure generation, and language and editorial revision. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. All numerical statements were regenerated from machine-readable artifacts, and all figures were rendered by deterministic plotting code rather than a generative image model.

References

[1] R. Rahul, S. Paliwal, M. Sharma, and L. Vig, *Automatic information extraction from piping and instrumentation diagrams*, arXiv:1901.11383, 2019.

[2] S. Paliwal, A. Jain, M. Sharma, and L. Vig, *Digitize-PID: Automatic digitization of piping and instrumentation diagrams*, arXiv:2109.03794, 2021.

[3] H. Kim, W. Lee, M. Kim, Y. Moon, T. Lee, M. Cho, and D. Mun, *Deep-learning-based recognition of symbols and texts at an industrially applicable level from images of high-density piping and instrumentation diagrams*, Expert Systems with Applications 183 (2021) 115337. doi:10.1016/j.eswa.2021.115337.

[4] G. Su, S. Zhao, T. Li, S. Liu, Y. Li, G. Zhao, and Z. Li, *Image format pipeline and instrument diagram recognition method based on deep learning*, Biomimetic Intelligence and Robotics 4(1) (2024) 100142. doi:10.1016/j.birob.2023.100142.

[5] V. Shteriyanov, R. Dzhusupova, J. Bosch, and H. Holmström Olsson, *From text to meaning: Semantic interpretation of non-standardized metadata in piping and instrumentation diagrams*, Computers & Chemical Engineering 204 (2026) 109436. doi:10.1016/j.compchemeng.2025.109436.

[6] M. Gupta, C. Wei, T. Czerniawski, and R. Eiris, *PIDQA—Question answering on piping and instrumentation diagrams*, Machine Learning and Knowledge Extraction 7(2) (2025) 39. doi:10.3390/make7020039.

[7] A. A. Alimin, D. P. Goldstein, L. Schulze Balhorn, and A. M. Schweidtmann, *Talking like Piping and Instrumentation Diagrams (P&IDs)*, Systems and Control Transactions (2025). doi:10.69997/sct.159477; arXiv:2502.18928.

[8] A. A. Alimin and A. M. Schweidtmann, *GraphRAG for Engineering Diagrams: ChatP&ID Enables LLM Interaction with P&IDs*, arXiv:2603.22528, 2026.

[9] B. Zhu, S. Duong, J. Vyas, and M. Mercangöz, *From P&ID Drawings to Process Graphs: A Multimodal Language Model Approach*, arXiv:2607.19568, 2026.

[10] K. Geras and C. Sutton, *Multiple-source cross-validation*, Proceedings of the 30th International Conference on Machine Learning, PMLR 28(3) (2013) 1292–1300.

[11] S. Kapoor and A. Narayanan, *Leakage and the reproducibility crisis in machine-learning-based science*, Patterns 4 (2023) 100804. doi:10.1016/j.patter.2023.100804.

[12] M. Mathew, D. Karatzas, and C. V. Jawahar, *DocVQA: A dataset for VQA on document images*, Proceedings of WACV (2021) 2200–2209.

[13] A. Masry, D. X. Long, J. Q. Tan, S. Joty, and E. Hoque, *ChartQA: A benchmark for question answering about charts with visual and logical reasoning*, Findings of ACL (2022) 2263–2279. doi:10.18653/v1/2022.findings-acl.177.

[14] A. Nourbakhsh, S. Parekh, P. Shetty, Z. Jin, S. Shah, and C. Rose, *Where is this coming from? Making groundedness count in the evaluation of Document VQA models*, Findings of NAACL (2025) 5341–5361. doi:10.18653/v1/2025.findings-naacl.295.

[15] L. Chen, X. Yan, J. Xiao, H. Zhang, S. Pu, and Y. Zhuang, *Counterfactual samples synthesizing for robust visual question answering*, Proceedings of CVPR (2020) 10800–10809. doi:10.1109/CVPR42600.2020.01081.

[16] Y. Li, Y. Du, K. Zhou, J. Wang, X. Zhao, and J.-R. Wen, *Evaluating object hallucination in large vision-language models*, Proceedings of EMNLP (2023) 292–305. https://doi.org/10.18653/v1/2023.emnlp-main.20.

[17] S. Leng, H. Zhang, G. Chen, X. Li, S. Lu, C. Miao, and L. Bing, *Mitigating object hallucinations in large vision-language models through visual contrastive decoding*, Proceedings of CVPR (2024) 13872–13882.

[18] Y. Du et al., *PP-OCR: A practical ultra lightweight OCR system*, arXiv:2009.09941, 2020.

[19] S. Bai et al., *Qwen3-VL technical report*, arXiv:2511.21631, 2025.

[20] W. Wang et al., *InternVL3.5: Advancing open-source multimodal models in versatility, reasoning, and efficiency*, arXiv:2508.18265, 2025.

[21] DEXPI e.V., *Public Example PIDs*, GitLab repository, commit a23d61e2e089eb2ca464cd552f9ae580a2785963, CC BY 4.0, 2024. https://gitlab.com/dexpi/TrainingTestCases.